



BIO411/BIO629 End semester Exam (2025) IISER Mohali (Part A)

Date: 21/11/2025

Duration: 30 min

Maximum marks: 20

Instructions:

a. All questions are compulsory. Each question carries 1 mark.

1. In Multiple Sequence Alignment (MSA) of diverse protein family sequences, which method is most likely to produce the most accurate alignment:
 - a. Chou Fasman
 - b. FASTA
 - c. MUSCLE**
 - d. Needleman-Wunsch
2. Highly conserved glycine residue at a specific position in a protein family is most likely due to:
 - a. Glycine is the most hydrophobic amino acid.
 - b. It is located on the protein's flexible, unstructured surface.
 - c. It lies in turn or in region requiring minimal steric hindrance.**
 - d. It has side chain interacting making important interactions.
3. 'HETATM' in Protein Data Bank (PDB) file format contains:
 - a. X, Y, and Z coordinates of each protein atom.
 - b. Residue name, B-factor, occupancy of each atom.
 - c. Defines interacting atoms for each non-protein atom.
 - d. X, Y, and Z coordinate of each defined non-protein atom.**
4. A high Debye-Waller factor (temperature factor) for a specific loop region in a crystallographic structure generally indicates:
 - a. It is highly hydrophobic.
 - b. The atoms in that region have high disorder, or high thermal motion.**
 - c. The region is involved in the active site of an enzyme.
 - d. Measure of error in determining structure.
5. A tool for secondary structure prediction of a protein sequence you can use:
 - a. BLASTP against the PDB
 - b. PSIPRED**
 - c. DSSP
 - d. STRIDE
6. A technique most suitable for determining 3D structure variability of a small protein (< 20 kDa) in solution at atomic resolution?
 - a. X-ray Crystallography
 - b. Cryo-Electron Microscopy (Cryo-EM)
 - c. Nuclear Magnetic Resonance (NMR) Spectroscopy**
 - d. Circular Dichroism (CD) Spectroscopy
7. β -turn often involves a residue with a positive phi angle. Amino acid most frequently found at this position is:
 - a. Tryptophan
 - b. Glycine**
 - c. Cysteine
 - d. Methionine
8. Several non-glycine residues in the "left-handed alpha-helix" region of a Ramachandran plot is typically a sign of:
 - a. A novel protein fold.
 - b. Excellent model quality and refinement.
 - c. Potential errors in the structure determination or refinement.**
 - d. A protein composed entirely of proline residues.
9. Proline rarely occurs in the middle of a standard alpha-helix because it's:
 - a. Side chain donates a hydrogen bond that disrupts the helix.
 - b. Lack of an amide hydrogen necessary for intra-helical hydrogen bond.**
 - c. Carries a charge that repels the helix macrodipole.

- d. Too flexible and destabilizes the helical stack.
10. Of these phylogenetic reconstruction, which follows molecular clock hypothesis:
- Neighbour-joining method
 - Fitch-Margoliash method
 - UPGMA**
 - ClustalW
11. For which amino acid χ_1 torsion angle is undefined:
- Only glycine
 - Only alanine
 - Proline, alanine and glycine
 - Alanine and glycine**
12. The region within $\pm 10^\circ$ of ϕ angle is primarily disallowed due to steric hindrance of atoms:
- C^β and H_{i+1}
 - O_{i-1} and C^β
 - O_{i-1} and C**
 - N and O_{i-1}
13. A protein X of length 150 residues has approximate diameter of 30 Å (assume spherical shape). In this protein, the longest possible ideal α -helix will have **20** number of residues.
14. In protein structural classification (SCOP), two proteins are related at the level of 'Fold' if these structures share **major secondary structure in same arrangement with topological connections**. The equivalent of *Fold* in CATH is referred to as **topology**.
15. In protein three-dimensional (3D) structure modeling/prediction context, *Template* refers to **protein with experimentally known structure used for modeling target sequences**.
16. The best structural homologue of pdb id 5pntA is **1got** from the data given below (sequences for structures differ from each other):

Pdb id	RMSD	TM-score
1a6m	0.9 Å	0.920
1got	1.2 Å	0.950
1goy	3.0 Å	0.416
1qra	2.0 Å	0.617
1dra	4.3 Å	0.324

17. Solvent accessible surface area (SASA) is defined as the **surface carved by locus of the centre of a probe sphere (representing the solvent molecule) as it rolls over the Van der Waals surface of the protein allowing maximum permitted contact**. Program **NACCESS/GETAREA** can be used to calculate SASA.
18. Gene Ontology (GO) is **controlled vocabulary of standardized terms (framework) to describe protein functions**. Three ontologies or aspects defined for GO are **Molecular function, cellular component and Biological process**.
19. Inverse protein folding problem is defined **as finding all possible sequences that can adopt a given fold**.
20. Sequence motif is defined as **a short, conserved pattern of nucleotides or amino acids that is associated with a particular biological function**.